

Hybrid Scripts Guide

This guide covers PowerShell scripts for managing users in a hybrid environment. These hybrid scripts have been created/modified by Nexus Open Systems for Cardo Group Ltd.

Any questions, please reach out to Nexus Service Desk - support@nexusos.co.uk

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Prerequisites

- Please run these scripts on one of Cardo's Domain Controllers or MGMT-01
 - User must have appropriate permissions (AD admin for on-premises, Microsoft Graph scopes for cloud operations)
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Organizational Unit (OU) Management

CreateCompanyOUs.ps1

Purpose: Create standard OU structure for a new company organization. This script ensures the creation of OUs for any newly acquired company to fit the existing OU structure on the domain controller.

Structure Created:

```
Company Name
├── Operative
├── Non-operative
│   ├── Finance
│   ├── Maintenance
│   ├── IT
│   └── Procurement
```

Usage:

```
.\CreateCompanyOUs.ps1
# Prompts for company name interactively (please ensure that the company name is
entered properly)
```

MoveExistingUsersOU.ps1

Purpose: Move existing users to different organizational units from CSV.

Inputs: CSV with user UPNs and target OU paths.

User Management Scripts

For all user management scripts, please start with the UserImportTemplate.xlsx file where all user data should be added. Next, save this file as csv. If saved as UserImport.csv, the scripts will automatically detect the file, otherwise, when a script runs, user will be prompted to enter the path of the csv file.

CreateNewUser.ps1

Purpose: Create new AD user accounts from CSV file.

Inputs: UserImport.csv with columns:

- FirstName, LastName, DisplayName, OtherName
- SamAccountName, UserPrincipalName, Title, Department, Company
- Manager, OU path, Description, Office, OfficePhone, EmailAddress, MobilePhone
- AccountPassword, AccountStatus (Enabled/Disabled), ChangePasswordAtLogon
- IsFieldOperative, D365FieldService

Features:

- Sets extension attributes for operative status
- Supports setting manager relationships
- Configures organizational unit placement
- Operative status set in the csv file will determine whether a user is assigned E3 or F3 license bundles

Usage:

```
.\CreateNewUser.ps1  
# Looks for UserImport.csv in script directory or prompts for path
```

AD_ExportAllUsers.ps1

Purpose: Export all on-premises AD users with comprehensive properties to CSV.

Exports: SAM Account Name, UPN, DisplayName, Email, Phone, Address, Department, Title, Company, Manager, Extension Attributes (1-15), Status, Creation Date.

Usage:

```
.\AD_ExportAllUsers.ps1  
.\AD_ExportAllUsers.ps1 -OutputPath "C:\Exports\AllADUsers.csv"
```

Output: CSV file (default: AD_Users_Export.csv)

Set_Operative_Or_NonOperative.ps1

Purpose: Set field operative status via extension attributes from CSV.

Inputs: CSV with UserPrincipalName and operative status.

Contact Information Updates

Update_MobileNumber.ps1

Purpose: Update mobile phone numbers in AD from CSV.

Inputs: **UserImport.csv** with Mobile column.

Features: Validates phone format, logs all changes.

Update_Telephone.ps1

Purpose: Update office telephone numbers in AD from CSV.

Inputs: **UserImport.csv** with TelephoneNumber column.

Update_Fax_For_Direct-Dial.ps1

Purpose: Update fax/direct-dial numbers in AD.

Inputs: CSV with fax number information.

Remove_MobileNumber.ps1

Purpose: Remove mobile phone numbers from user accounts.

Usage: Typically used in leaver or data cleanup scenarios.

Update_Address.ps1

Purpose: Update address information (street, city, state, postal code) from CSV.

Inputs: CSV with address fields.

Update_CompanyOrLocation.ps1

Purpose: Update company and/or location fields from CSV.

Authentication Methods

Add_PhoneAuthenticationMethod.ps1

Purpose: Add phone authentication method for Entra ID users (MFA).

Inputs: CSV with columns:

- UserPrincipalName
- AuthenticationPhone (format: +44123456789)

Features: Connects to Microsoft Graph, validates phone format.

Usage:

```
.\Add_PhoneAuthenticationMethod.ps1
```

Get_PhoneAuthenticationMethod.ps1

Purpose: Retrieve phone authentication methods currently assigned to users.

Usage: Useful for auditing MFA setup.

Remove_PhoneAuthenticationMethod.ps1

Purpose: Remove phone authentication methods from users.

Usage: Used when removing MFA or during leaver processing.

Password Management

Set_Passwords.ps1

Purpose: Set initial passwords for users in AD from CSV.

Inputs: **UserImport.csv** with Password column.

Features:

- Sets AD password directly
- Supports "ChangePasswordAtLogon" flag
- Logs all operations

Usage:

```
.\Set_Passwords.ps1
```

License Management

Set_E3NoTeams_License.ps1

Purpose: Assign Microsoft 365 E3 license (without Teams) to users from CSV.

Inputs: CSV with user identifiers.

Notes: Uses extension attribute 3.

Set_D365_Field_License.ps1

Purpose: Assign Dynamics 365 Field Service license from CSV.

Inputs: CSV with user identifiers and license configuration.

Notes: Uses extension attribute 2

SMTP & Email Configuration

Set_SMTP_Address.ps1

Purpose: Configure primary SMTP address for users from CSV.

Inputs: CSV with EmailAddress column.

Features: Updates both AD and Exchange Online attributes.

Leaver Processing

Leaver_Script.ps1

Purpose: Comprehensive off-board employee in hybrid environment.

Actions (performed in both AD and Entra ID):

1. Disable AD user account
2. Disable Entra ID user account
3. Convert mailbox to shared mailbox
4. Remove all Entra ID licenses
5. Clear extension attributes (1-15)
6. Remove user from all Entra ID security groups
7. Move disabled account to _Leavers OU
8. Log all actions

Inputs: `UserImport.csv` with AccountStatus set to "Disabled"

Usage:

```
.\Leaver_Script.ps1  
.\Leaver_Script.ps1 -CsvPath "C:\Path\To\UserImport.csv"
```

Output: Timestamped log file with all actions taken.

Notes: Requires permissions in both on-premises AD and cloud (Entra ID, Exchange Online).

CSV File Format

Most scripts use a CSV file with the following general columns (not all required for every script):

```
FirstName,LastName,DisplayName,UserLogonName,UserPrincipalName>Password,AccountSta  
tus,JobTitle,Department,Company,TelephoneNumber,Mobile,E-  
mail,ManagerEmail,IsFieldOperative,D365FieldService
```

Column Details:

- **UserLogonName:** SAM Account Name (no @domain)
 - **UserPrincipalName:** Full UPN (user@domain)
 - **AccountStatus:** "Enabled" or "Disabled"
 - **IsFieldOperative:** "Yes" or "No"
 - **D365FieldService:** License assignment flag
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Important Notes

- **Hybrid Sync:** Changes must be synced from AD to Entra ID via Azure AD Connect
 - **Extension Attributes:** Values in extension attributes (1-15) are synced from AD to Entra ID
 - **Leaver OU:** Ensure a "_Leavers" OU exists for the leaver script to move disabled accounts
 - **Permissions:** Ensure service account has sufficient rights in both AD and cloud environments
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Note: All scripts are designed for hybrid environments where on-premises AD is synced to Entra ID via Azure AD Connect. Some operations affect both on-premises and cloud resources.